American Computer Science League

2024-2025 ● Contest 2: Short Problems ● Elementary Division Solutions

1. Prefix/Infix/Postfix

In evaluating a prefix expression, use 2 operands preceded by an operator from left to right:
* + 3 9 − 8 2
= * (3 + 9) (8 − 2)
= (3 + 9) * (8 − 2)
= 12 * 6
= 72

A. 72

2. Prefix/Infix/Postfix

In evaluating a postfix expression, use 2 operands followed by an operator from left to right:
9 8 − 5 2 ^ + 8 5 + /
= (9 − 8) (5 ^ 2) + (8 + 5) /
= ((9 − 8) + (5 ^ 2)) / (8 + 5)
= (1 + 25) / 13
= 26 / 13
= 2

D. 2

3. Prefix/Infix/Postfix

To convert an infix expression to prefix, use the order of operation:

$$\frac{5 + 4^2}{9 - 3*2}$$

= (5 + 4 ^ 2) / (9 − 3 * 2)
= (5 + (^ 4 2)) / (9 − (* 3 2))
= (+ 5 ^ 4 2) / (− 9 * 3 2)
= / + 5 ^ 4 2 − 9 * 3 2

E. / + 5 ^ 4 2 − 9 * 3 2

4. Prefix/Infix/Postfix To convert an infix expression to postfix, use the order of operations: 3 * 5 − 9 / 3 ^ 2 + 7 = (3 * 5) − (9 / (3 ^ 2)) + 7 = (3 5 *) − (9 / (3 2 ^)) + 7 = (3 5 *) − (9 3 2 ^ −) + 7 = (3 5 * 9 3 2 ^ − −) + 7 = 3 5 * 9 3 2 ^ / − 7 +	B. 3 5 * 9 3 2 ^ / − 7 +
5. Prefix/Infix/Postfix Each of the following expressions is evaluated as follows: a. 8 4 − 2 / = (8 − 4) / 2 = 4 / 2 = 2 b. 8 4 2 − / = 8 / (4 − 2) = 8 / 2 = 4 c. − / 8 4 2 = (8 / 4) − 2 = 2 − 2 = 0 d. − 8 / 4 2 = 8 − (4 / 2) = 8 − 2 = 6 Therefore, the order from lowest value to highest value is c, a, b, d.	C. c, a, b, d
6. Prefix/Infix/Postfix To convert a postfix expression to prefix, put the operator before the operands as opposed to after the operands: 5 3 * 6 2 / + 2 3 ^ − 5 / = (5 3 *) (6 2 /) + (2 3 ^) − 5 / = ((((5 3 *) (6 2 /) +) (2 3 ^) −) 5 /) = (− (+ * 5 3 / 6 2) (^ 2 3)) 5 / = / − + * 5 3 / 6 2 ^ 2 3 5	E. / − + * 5 3 / 6 2 ^ 2 3 5